

Executive Summary: Stolen Artworks Intelligence Dashboard

The Stolen Artworks Intelligence Dashboard is a database-backed web application designed to help users explore patterns in major stolen-artwork cases. The project uses a Supabase PostgreSQL database to store structured case records and a React/Vite frontend deployed on Netlify to visualize the data through key performance indicators, charts, filters, and a searchable case table. The theme is art crime, with records covering famous missing and recovered artworks, including paintings, sculptures, manuscripts, and cultural objects.

The database contains fields such as artwork title, artist, category, theft location, institution, theft year, recovery year, recovery status, estimated value, risk score, and public source link. This structure allows the dashboard to answer practical questions such as which cases remain missing, which countries appear most often in the dataset, which categories are most exposed, and how recovery outcomes differ across historical periods. The dashboard includes KPI cards for total cases, missing cases, recovered cases, recovery rate, estimated value, and average recovery time. It also includes timeline, status, category, country, and value-by-year visualizations.

The application is intentionally built for classroom demonstration and executive communication. The interface is visual and easy to explain, while the database design demonstrates core full-stack concepts: relational schema design, SQL seeding, row-level read policy, environment variables, frontend data fetching, filtering, and deployment. Unlike earlier versions, this stable version does not use frontend ETL tokens or protected refresh functions. The Refresh view button simply reloads the current Supabase database records and avoids token-related 401/500 errors.

The dashboard can support museum-security discussions, cultural-heritage risk analysis, and data-literacy teaching. Its main limitation is that it uses a curated educational dataset rather than an official law-enforcement feed. Future improvements could include authenticated admin editing, verified API ingestion from public registries, geographic mapping, and richer provenance fields. Overall, the project meets the assignment goal by delivering a functional deployed web app, a structured database, visual analytics, and an executive summary.